

R360 Product Requirements Document (PRD)

1. Executive Summary

R360 is a centralized enterprise platform designed to replace spreadsheet-based resource planning and provide a single source of truth for **projects, skills, resource allocation, time tracking, and capacity planning**.

The system enables **project-driven, skill-based resource allocation, role-based effort forecasting, weekly time accounting via project codes, and robust reporting** to support delivery accuracy, utilization tracking, and leadership decision-making.

3. Goals & Objectives

- Eliminate manual Excel-based resource planning
 - Enable skill- and availability-driven allocation
 - Forecast project effort accurately by role
 - Capture actual effort weekly using standardized time codes
 - Provide reliable utilization and capacity reports
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4. In-Scope / Out-of-Scope

In Scope

- Project management & capacity planning
- Skills & role-based requirements
- Resource allocation
- Weekly time entry (project, leave, holiday)
- Reporting & utilization

Out of Scope (Phase 1)

- Payroll integration
 - Billing & invoicing
 - Leave balance management
 - AI recommendations
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5. User Roles & Access Control

Role	Description
Admin	Full system control
Project Manager (PM)	Manages assigned projects
Employee	Time entry & self-view
Leadership	Read-only dashboards

Global Access Matrix

Module	Admin	PM	Employee	Leadership
Project Master	Full	Own Projects	View	View
Skills Master	Full	View	View/Edit Self	View
Resource Allocation	Full	Own Projects	View	View
Time Entry	Full	View/Approve	Enter	View
Reports	Full	Project-level	Self	Full

6. Skills & Role Framework

6.1 Skills Master

Field	Description
Skill Name	Unique technology/skill
Skill Category	Backend, Frontend, DevOps, etc.
Status	Active / Deprecated

- Managed by Admin only
- Used across projects and employees

6.2 Employee Skills

Each employee skill is classified as:

- **Primary Skill** (core expertise)
- **Secondary Skill** (supporting skill)

Field	Description
Skill Name	From Skills Master
Skill Type	Primary / Secondary
Skill Level	Beginner / Intermediate / Expert
Experience (Years)	Optional

Rules:

- At least one Primary Skill mandatory
- Primary skills are prioritized during allocation
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6.3 Role Labels (Non-blocking)

Roles act as **descriptive labels only**.

Examples:

- Backend
- Frontend
- Full Stack
- Architect

Rules:

- Role does not override skill matching
- Used as secondary ranking/filtering criterion

7. Project Management (Single Source of Truth)

7.1 Project Creation & Core Fields

Access: Admin, PM

Field	Mandatory	Description
Project Name	Yes	Unique name
Project Code	Auto	Used for time tracking
Project Owner (PM)	Yes	Responsible owner
Start Date	Yes	Project start
End Date	Yes	Project end
Status	Auto	Active / Completed
Priority	Yes	High / Medium / Low
Billing Type	No	Billable / Non-billable

7.2 Project Capacity & Effort Planning (Critical)

Field	Mandatory	Description
Projected Total Hours	Yes	Total effort required
Delivery Model	No	Fixed / T&M

7.3 Project Skill & Role Requirements

Defined **before allocation**.

Field	Description
Skill / Tech Stack	Required skill
Minimum Skill Level	Expected proficiency
Required Headcount	Number of people
Required Days per Person	Planned days
Role Label	Preferred role

Derived:

None

$\text{Projected Hours per Role} = \text{Headcount} \times \text{Days} \times \text{Hours/Day}$

8. Resource Allocation (Project-Centric)

8.1 Allocation Entry Point

Path: **Projects** → **Project** → **Resource Allocation**

Access: Admin (Full), PM (Own projects)

8.2 Employee Search & Matching Logic

Employees are ranked using the following order:

1. **Primary Skill Match** (first priority)
2. Skill Level match
3. Role Label match
4. Availability % (descending)
5. Experience (tie-breaker)

Availability Calculation:

None

$\text{Availability \%} = 100 - \text{Current Allocation \%}$

Employees with **100% allocation appear last.**

8.3 Allocation Fields

Field	Description
Employee	Selected from ranked list
Skill Used	Primary/Secondary
Allocation %	1–100
Allocation Start	$\geq$ Project Start
Allocation End	$\leq$ Project End
Forecasted Hours / Week	Auto-derived

Rules:

- PM cannot over-allocate (>100%)
- Admin override allowed (logged)

9. Time Accounting & Project Codes

9.1 Time Code Types

Type	Description
Project Code	Auto-created per project
Leave Code	Planned/Sick leave
Holiday Code	System-generated

9.2 Weekly Time Entry

Access: Employee (Enter), PM (View), Admin (Edit)

Field	Description
Week	Auto
Time Code	Project / Leave / Holiday
Project	Auto (if project code)
Role	From allocation
Hours	Numeric
Comments	Optional

Rules:

- Only allocated projects selectable
- Total weekly hours capped (configurable)
- Holiday entries auto-populated

10. Forecast vs Actual Tracking

Tracked at:

- Employee level
- Role level
- Project level

Metrics:

- Planned (Role-based)
- Forecasted (Allocation-based)
- Actual (Time entry)
- Variance

11. Reporting & Dashboards

Core Reports

Report	Audience
Employee Utilization	Admin, Leadership
Project Capacity vs Actual	Admin, PM
Role-wise Effort	Admin, Leadership
Forecast vs Actual	Admin, PM

Utilization Formula:

None

$$\text{Utilization \%} = \text{Project Hours} \div \text{Available Hours}$$

Leave & Holiday excluded.

13. Risks & Mitigation

Risk	Mitigation
Poor time entry adoption	Weekly reminders & lock dates
Skill data inconsistency	Admin governance
Over-allocation	Hard PM limits, Admin override only

16. Data Model / Entity Relationship (ER) Diagram

16.1 Core Entities Overview

Master Entities

- Employee
- Skill
- Role
- Project
- Time Code

Transactional Entities

- Employee Skill
- Project Skill Requirement
- Project Role Effort
- Project Allocation
- Time Entry

16.2 Entity Definitions

16.2.1 Employee

Field	Type	Notes
employee_id (PK)	UUID	Primary key
first_name	String	
last_name	String	
email	String	Unique
role_label_id (FK)	UUID	Backend / Frontend etc.
department	String	
position	String	
status	Enum	Active / Inactive
join_date	Date	
created_at	Timestamp	

16.2.2 Role

Field	Type	Notes
role_label_id (PK)	UUID	
role_name	String	Backend, Frontend, etc.
status	Enum	Active / Inactive

16.2.3 Skill

Field	Type	Notes
skill_id (PK)	UUID	
skill_name	String	Unique

skill_category	String	Backend, Frontend, DevOps
status	Enum	Active / Deprecated

16.2.4 Employee Skill

Field	Type	Notes
emp_skill_id (PK)	UUID	
employee_id (FK)	UUID	→ Employee
skill_id (FK)	UUID	→ Skill
skill_type	Enum	Primary / Secondary
skill_level	Enum	Beginner / Intermediate / Expert
experience_years	Number	Optional

Relationship:

- Employee (1) — (M) Employee Skill
 - Skill (1) — (M) Employee Skill
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16.2.5 Project

Field	Type	Notes
project_id (PK)	UUID	
project_name	String	Unique

project_code	String	Auto-generated
project_owner_id (FK)	UUID	→ Employee
start_date	Date	
end_date	Date	
status	Enum	Active / Completed
priority	Enum	High / Medium / Low
billing_type	Enum	Billable / Non-billable
delivery_model	Enum	Fixed / T&M
projected_total_hours	Number	

16.2.6 Project Skill Requirement

Field	Type	Notes
proj_skill_req_id (PK)	UUID	
project_id (FK)	UUID	→ Project
skill_id (FK)	UUID	→ Skill
min_skill_level	Enum	Beginner+
required_headcount	Number	
required_days	Number	Per person

Relationship:

- Project (1) — (M) Project Skill Requirement
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16.2.7 Project Role Effort

Field	Type	Notes
proj_role_effort_id (PK)	UUID	

project_id (FK)	UUID	→ Project
role_label_id (FK)	UUID	→ Role
required_headcount	Number	
required_days	Number	Per person
hours_per_day	Number	Default 8
projected_role_hours	Number	Derived

16.2.8 Project Allocation

Field	Type	Notes
allocation_id (PK)	UUID	
project_id (FK)	UUID	→ Project
employee_id (FK)	UUID	→ Employee
skill_id (FK)	UUID	Skill used
allocation_percentage	Number	1–100
allocation_start_date	Date	
allocation_end_date	Date	
forecast_hours_per_week	Number	Derived

Relationships:

- Project (1) — (M) Project Allocation
- Employee (1) — (M) Project Allocation

16.2.9 Time Code

Field	Type	Notes
time_code_id (PK)	UUID	
code	String	PRJ-XXX, LV-PL
type	Enum	Project / Leave / Holiday
project_id (FK)	UUID	Nullable
status	Enum	Active / Inactive

16.2.10 Time Entry

Field	Type	Notes
time_entry_id (PK)	UUID	
employee_id (FK)	UUID	→ Employee
time_code_id (FK)	UUID	→ Time Code
project_id (FK)	UUID	Nullable
role_label_id (FK)	UUID	Derived
week_start_date	Date	
hours	Number	
comments	String	Optional
created_at	Timestamp	

16.3 High-Level ER Relationship Summary

None

```
Employee ——< EmployeeSkill >—— Skill
|
|——< ProjectAllocation >—— Project ——<
ProjectSkillRequirement >—— Skill
|
|                                     |——< ProjectRoleEffort >—— Role
|
|——< TimeEntry >—— TimeCode —— Project
```

16.4 Reporting Alignment

This data model supports:

- Skill-based allocation analytics
- Role-wise effort planning vs actuals
- Employee & project utilization
- Forecast vs actual variance
- Audit & traceability